CARDIOVASCULAR DISEASE CLASSIFICATION IN NUMPY

Classifying reported cardiovascular conditions

Classify whether respondents to the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a U.S. health survey, reported myocardial infarction or coronary heart disease.

328,135
labeled respondents

321
survey predictors

109,379
unlabeled test respondents

NumPy
models and preprocessing, plus the Python standard library

Final submission: AIcrowd F1 0.455 · 3rd on the AIcrowd leaderboard as **anto_arbi** (14 October 2026)

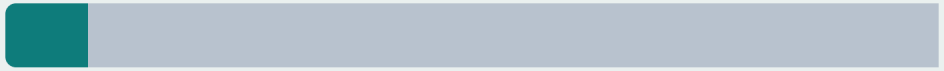
Antonio Arbizzani Tonelli · EPFL Machine Learning CS-433

An uncommon positive class and substantial missingness

CLASS BALANCE · RESPONDENTS

8.83%

28,975 of 328,135 training respondents are positive.



8.83% positive 91.17% negative

MISSINGNESS · PREDICTOR CELLS

44.79%

of raw training predictor cells are blank. Many blanks reflect questionnaire branching rather than a zero quantity.

Why F1 guides selection

Predicting every respondent as negative gives 91.17% training accuracy and F1 = 0. F1 is used because positive recovery and false positives both matter.

Source: evidence/METRICS.json (dataset), project/docs/DATA.md

Response codes need their survey meanings

Codes are interpreted separately for each variable: a finite number can represent a quantity, a category or a nonresponse.

Survey code	Meaning in the codebook	Becomes
ALCDAY5 = 888	No alcohol consumption	0
PHYSHLTH = 88	Zero physically unhealthy days	0
Documented unknown or refused codes	Nonresponse, defined per variable	missing (NaN)

1,095,421
nonresponse codes mapped to missing

899,687
zero-quantity codes mapped to 0

Historical cleaning audit across 262,508 development rows and 295 retained columns. Transformation counts, not evidence of a performance gain. No universal rule applies to codes such as 7, 9, 88 or 888.

Source: phase14_codebook_corrections_only.json, phase33_preprocessing_counts.json

The final representation adds diet exclusions and restricted imputation

Diet exclusions

Exclude six raw diet questions that mix daily, weekly and monthly codes. Keep the derived daily-frequency columns already in the data.

Restricted imputation

Fill only specified unknown or refused answers and blank height/weight; derive missing BMI. Imputers learn from training features, without disease labels, inside each training partition.

297 features

289 source columns + 8 missing-state flags

Questionnaire-branching blanks stay missing, and the original missing-state indicators are preserved.

Both changes start from the historical codebook cleaning. Not every missing value is filled, and the daily-frequency columns are not newly calculated by this pipeline.

Source: combined18_24_preprocessing.json, conditional_imputation.py, project/docs/DATA.md

Histogram gradient boosting in NumPy

1 Bin

Up to 64 quantile bins per feature, learned from training rows

2 Fit

Trees fitted with Newton updates on logistic loss, depth up to 5

3 Sum

200 tree outputs added with learning rate 0.05

4 Decide

Positive when $p \geq$ the saved threshold 0.2181

MISSING VALUES

Each split learns which branch receives missing responses.

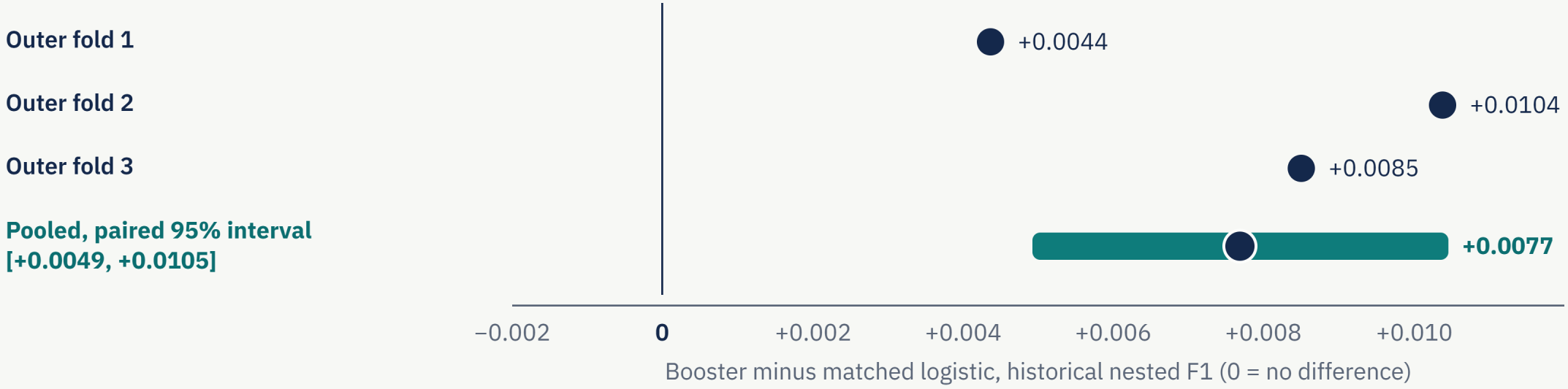
SOURCE LIMITATION

The preserved historical source omits some observed-versus-missing split candidates; a corrected search was evaluated separately.

The final pipeline uses the byte-identical historical booster source. Remaining parameters are in Appendix A.

Source: src/numpy_boosting_phase14.py, configs/final_model.json, FINAL_REPORT.md

The historical booster beat matched logistic regression

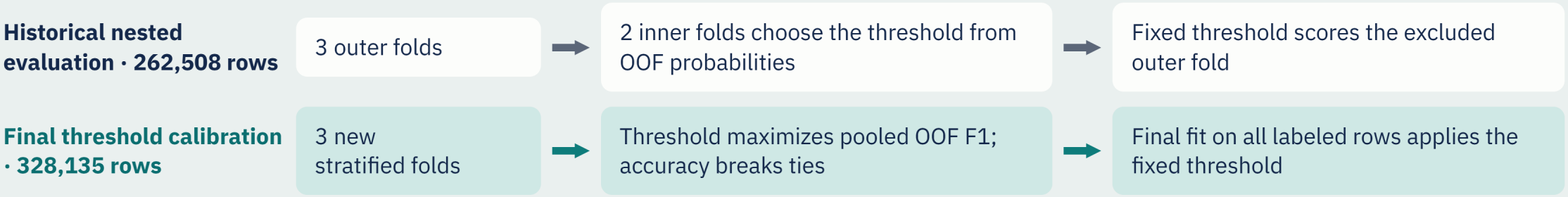


Pooled nested F1: reference booster **0.442155**, matched logistic **0.434467**. Same cleaning, 295 source variables, development rows and folds; logistic regression has its own fitted filling, scaling and missingness flags, so this compares complete pipelines. The interval is conditional on the fitted models and thresholds.

This historical comparison does not evaluate the final combined pipeline.

Source: phase38_matched_logistic_summary.md, assets/chart_data/model_comparison.csv

Evaluation and threshold calibration are separate



Pipeline and protocol	F1	Interpretation
Historical reference, nested development	0.442155	Earlier pipeline, excluded outer folds
Final combination, OOF threshold calibration	0.443882	Same labels select and score the threshold
Final combination, AIcrowd feedback	0.455	Author-reported, used in selection
Final combination, nested development	n/a	No completed estimate

Preprocessing is learned inside each training partition. Calibration F1 is not an independent test, and this is threshold calibration, not a probability-calibration model. The values are not successive improvements.

Source: final_phase14_validation.json, final_combined18_24_submission.json, final_model.json

Challenge feedback selected the final combination

Completed candidate	Displayed F1	Accuracy
Historical reference	0.452	0.878
Diet representation only	0.453	0.882
Restricted imputation only	0.454	0.880
Leaf L2 = 5 only	0.453	0.883
Diet + restricted imputation, L2 = 1 (selected)	0.455	0.882
Diet + restricted imputation, L2 = 5	0.453	0.883

+0.003

displayed F1 over the historical submission; not a statistically reliable improvement

F1 decided the choice even where another candidate has higher accuracy. Each candidate uses its own OOF-selected threshold.

Locally, diet only (0.4419) and imputation only (0.4418) did not exceed the reference (0.4422).

Author-reported AIcrowd display scores, rounded to three decimals: historical reference on 12 October 2026, other candidates on 13 October 2026. Direct submission URLs and screenshots were not supplied. Repeated feedback informed selection.

Source: aicrowd_progression.csv, evidence/aicrowd/, final_combined18_24_submission.json

Post-event information limits the interpretation

Event
Heart attack

→

Collected after the event
HAREHAB1: rehabilitation

→

Target, _MICHD
Reported MI or coronary disease

620 / 620
development respondents with a recorded response are positive; the feature remains in the final pipeline

0.442155 → 0.431039
historical nested F1 without it; this ablation does not measure its effect in the final combination

The project classifies reported disease in a survey. It does not demonstrate prospective clinical prediction, and no temporal or external-population validation is established. Development folds and challenge feedback were consulted repeatedly, so model-selection optimism remains.

Source: feature_metadata.json, evidence/METRICS.json, FINAL_REPORT.md (HAREHAB1)

Reproduction and dated release evidence

From project/, with the documented environment and the official challenge CSVs:

```
python run.py --data-dir data/raw/dataset --output submission.csv
```

Entry point

Fits all labeled rows and uses the threshold stored in the selected configuration.

109,379 predictions

13,999 positive and 95,380 negative: prediction counts, not classification errors.

14 October 2026 checks

242 synthetic project tests and exact replay of the selected predictions; not new training.

Guides from the repository root: docs/RESULTS.md · project/README.md · evidence/RELEASE_20261014.json. Data, checkpoints and row-level predictions stay outside Git.

3rd on the AIcrowd leaderboard as anto_arbi on 14 October 2026: an author-reported snapshot, not a final placing.

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Appendix A · Selected recipe and threshold

Setting	Selected value
Identifier	combined18_24_l2_1
Model input	289 source variables + 8 missing-state flags
Trees · depth · learning rate	200 · 5 · 0.05
Leaf rows · bins · features	≥ 200 · up to 64 · 128 per node
Leaf L2 · model seed	1 · 20260920
Decision threshold	0.21811010373987366
Threshold selection	Pooled 3-fold full-training OOF F1
Fold seed · tie-break	20260919 · accuracy

Repository identifiers

Phase 14: historical booster source and codebook cleaning.

Phase 18: diet exclusions.

Phase 24: restricted conditional imputation.

The final recipe applies the two later preprocessing changes to the unchanged Phase 14 source.

The booster source is byte-identical to the historical reference; the changed preprocessing and the calibrated threshold define a new pipeline. Preserving the source does not transfer the historical evaluation to it.

Source: configs/final_model.json, combined18_24_preprocessing.json, final_combined18_24_submission.json

Appendix B1 · Historical nested comparisons

Historical pipeline	F1	Precision	Recall	AP
Ridge	0.419508	0.333029	0.566652	0.390283
Earlier logistic regression	0.425628	0.357279	0.526316	0.387745
Matched logistic regression	0.434467	0.367541	0.531191	0.420990
Reference booster	0.442155	0.367632	0.554573	0.428609
Corrected split search	0.440803	0.362001	0.563460	0.428781
MLP 64 → 32	0.424451	0.360722	0.515531	0.387782
MLP 128 → 64	0.419268	0.340005	0.546721	0.379380

All rows are historical nested pipelines (262,508 rows, 3 outer and 2 inner folds, preprocessing fitted in training partitions); none evaluates the final combination. Ridge and earlier logistic regression keep their historical median-fill representations. AP = average precision.

Source: assets/chart_data/model_comparison.csv, evidence/METRICS.json

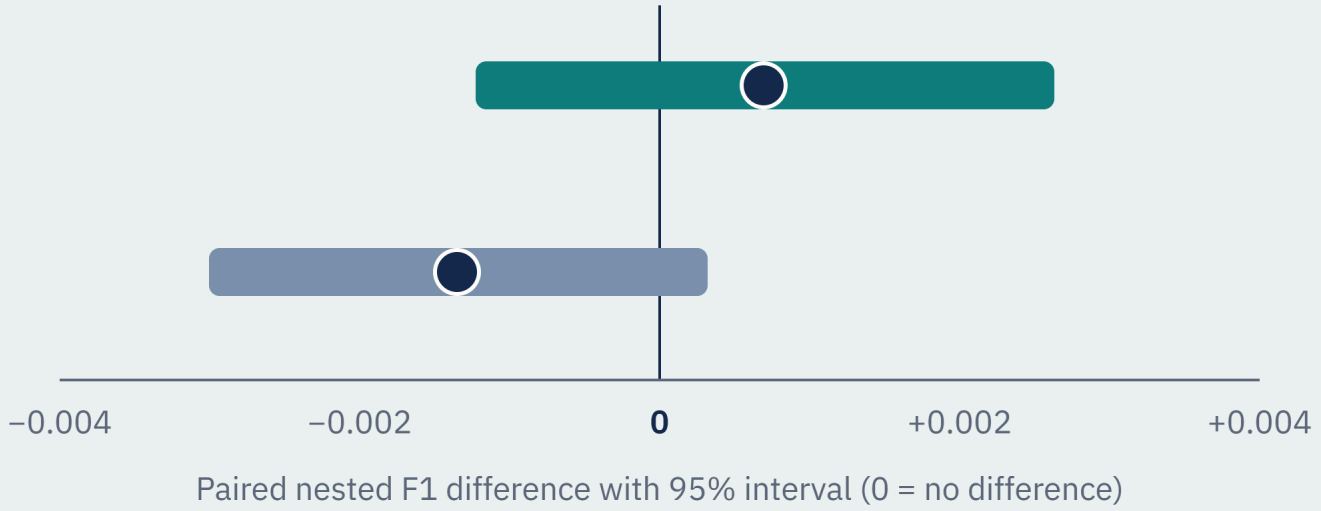
Appendix B2 · Two historical differences include zero

Cleaning and exclusions

cleaned – raw: +0.000698 [−0.001226, +0.002635]

Corrected split search

corrected – historical: −0.001352 [−0.003005, +0.000319]



Paired nested F1 difference with 95% interval (0 = no difference)

Both intervals are conditional on fixed fits and thresholds and exclude retraining and repeated selection. The cleaning benchmark combines recoding and exclusions. The final pipeline keeps the cleaning and the historical split search.

Source: phase33_raw_preprocessing_summary.md, phase34_missing_only_split_summary.md

Appendix C · Historical classification errors

Actual ↓ · Predicted →	Predicted positive	Predicted negative
Actual positive 23,180 respondents	**12,855** True positives	**10,325** False negatives
Actual negative 239,328 respondents	**22,112** False positives	**217,216** True negatives

0.367632
Precision: share of predicted positives that are reported cases

0.554573
Recall: share of reported cases that are found

Historical reference booster, nested development decisions on 262,508 rows. These counts are not an independent evaluation of the final model, and the final test prediction counts are not confusion-matrix counts.

Source: evidence/METRICS.json (Phase 14 reference, confusion_counts)